

Seeds, tiles and sophistication

how a random ReLU network strips randomness from its input

Working note II for the continuation-algebra programme
self-reviewed draft, not independently verified

3 October 2026

Abstract

We pursue the intuition that the input of a random ReLU network acts like a *seed*. The network stretches it into a long address (its activation pattern, i.e. the tile of input space it lands in) and a readout. Along the way it strips away most of the seed's randomness, in the sense of Kolmogorov sophistication: what survives in the expected output is *structure*, and the rest is *randomness* that only averages out. Four results make this precise.

(1) *Two-part codes in continuation algebras*. The centre of each level of the continuation algebra is the model and the arrows move the randomness. Block entropy splits as statistical complexity plus within-class randomness. For computable states the class label is an algorithmic two-part model, optimal along random sequences up to the model's deficiency. Readout means compress to one representative per class, at a cost bounded by $\sqrt{I(P;U \mid C)}/2$: the information the output keeps about *which* member of a class the input was.

(2) *Depth strips input randomness at a critical rate*. Under the weight average, the input-dependent fraction of the final output is $P(Z_L > 0)$ for a critical Galton–Watson process Z , about $9\pi^2/(2L^2)$. Activation-pattern bits of layer ℓ disagree between independent inputs with probability $\arccos(\varrho^{\circ(\ell-1)}(0))/\pi \approx 3/\ell$, and the rest are frozen by the weights. Both match simulations.

(3) *Inputs as seeds: an exact identity, a lower bound and an almost-optimal design*. For any finite set of input directions, the weight-averaged squared error of the induced estimator is a positive sum over moment orders with Galton–Watson weights. Antithetic pairs remove odd orders, and randomly rotated cross-polytope frames (3-designs) also remove order 2. Conversely, *no* nonnegatively weighted point set of size $M \ll n^2$ can remove even orders ≥ 4 . So frames are within a factor 2 of optimal, and no seed design can beat Monte Carlo by more than $P(Z_L > 0)/P(Z_L \in 2\mathbb{N}, Z_L \geq 4) \approx 3.3$ at depth 16. Predicted gains of frames over i.i.d. sampling are 4.33/2.42/1.66 at depths 4/8/16. At width 256 we observe 4.29/2.64/2.17, and at the challenge shape (1024×16) we observe 1.73 for frames and 1.08 for antithetic pairs (predicted 1.66 and 1.10).

(4) *The network's own tiling*. Activation patterns tile the input sphere hierarchically (refining with depth). The network is exactly linear on each tile, so within-tile randomness is stripped exactly and only tile barycentres survive. The nested tile expectations carry the same heat semigroup and continuation-class machinery as the Cantor hierarchy of note I. This is where the input enters the arrow algebra: as a point of the network-induced hull, addressed by its pattern string.

Contents

1	The picture	2
2	Two-part codes in continuation algebras	2

3	Depth strips input randomness at a critical rate	4
4	Inputs as seeds: identity, lower bound, near-optimal design	4
5	The network’s own tiling of the input	6
6	What this says about estimation	7
A	Numerical checks	7

1 The picture

Sophistication. A two-part description of an object x names a model $S \ni x$ and then x ’s index in S , so $K(x) \lesssim K(S) + \log |S|$. Models achieving near-equality are (algorithmic) sufficient statistics, and the least complex one measures *sophistication*: the structure, as opposed to the randomness (Koppel; Gács–Tromp–Vitányi; Vereshchagin–Vitányi). Its Shannon counterpart for processes is the causal state, the minimal sufficient statistic of the past for the future (Crutchfield–Young, Shalizi–Crutchfield). Note I showed that continuation classes *are* causal states.

PRG reading. A pseudorandom generator separates a short seed (randomness) from a fixed stretching map (structure), and its output fools a class of tests. Here:

- the *seed* is the network input (by homogeneity, its direction $\theta \in S^{n-1}$);
- the *stretch* is the activation-pattern address $\pi(\theta) \in \{0, 1\}^{nL}$ together with the output $F(\theta)$;
- the *tests* are the readouts, and the quantity of interest $\mathbb{E}F$ is pure structure.

Monte Carlo pays for the randomness. A mechanistic estimator tries to compute the structure directly. The question is how much of the seed the network actually uses, and at which “orders”.

Standing setting. Bias-free ReLU MLP, width n , depth L , ReLU after every layer, weights i.i.d. $N(0, s)$ with $s = 2/n$ (He), inputs $X \sim N(0, I_n)$ (WhestBench Phase 2: $n = 1024$, $L = 16$). Write $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ for the final post-ReLU map, F_j for its coordinates, σ for the uniform measure on S^{n-1} , and $c_n = \mathbb{E}|X| = \sqrt{2} \Gamma(\frac{n+1}{2}) / \Gamma(\frac{n}{2})$. Positive homogeneity gives the exact radial reduction

$$\mathbb{E}_\gamma F = c_n \mathbb{E}_\sigma F(\theta). \quad (1)$$

From note I we use the ReLU correlation map $\varrho(c) = \frac{c}{2} + \frac{1}{\pi}(\sqrt{1-c^2} + c \arcsin c)$ and its iterates $\varrho^{\circ D}(c) = \sum_r b_r^{(D)} c^r$. Here $b_r^{(D)} = P(Z_D = r)$ for the Galton–Watson process with offspring generating function ϱ , which is critical ($\varrho'(1) = 1$).

2 Two-part codes in continuation algebras

Let ν be a state on the binary hierarchy with continuation classes P_n , level algebras $B_n = \bigoplus_{C \in P_n} M_C$, prefix masses $\nu[p]$, class masses $\nu[C]$, and in-class weights $\rho_C = \text{diag}(\nu[p]/\nu[C])_{p \in C}$.

Definition 2.1. The *model algebra* at level n is the centre $Z_n = \text{span}\{e_C\}$ of B_n , and the *randomness* is the in-class position. $\Psi_n(x) = \sum_C \text{tr}(\rho_C x_C) e_C$ is the ν -preserving conditional expectation onto Z_n : “forget the randomness, keep the model”.

Proposition 2.2 (structure and randomness). (a) The arrows e_{pq} of B_n commute with Z_n ; their orbits on level- n prefixes are exactly the classes. The orbit space is the model and the position in an orbit is the randomness. (b) Block entropy splits as

$$H(P_n) = \underbrace{H(C_n)}_{\text{statistical complexity}} + \underbrace{\sum_C \nu[C] H(\rho_C)}_{\text{in-class randomness}},$$

i.e. $S(\nu|_{B_n}) = S(\nu|_{Z_n}) + \sum_C \nu[C] S(\rho_C)$ for the restricted states. (c) For $g \in Z_n$ (arrow-invariant readouts), $\nu(g) = \sum_C \nu[C] g_C$. Any β with $\beta[C] = \nu[C]$ for all C gives $\beta(g) = \nu(g)$: such readouts are fooled by any seed that reproduces only the class law.

Proof. (a) e_{pq} lies in the block M_C , which commutes with the centre, and e_{pq} maps p to q within C . (b) The restriction $\nu|_{B_n}$ is diagonal with entries $\nu[p] = \nu[C] \rho_C(p)$, so this is the chain rule. (c) Immediate. $\square$

Proposition 2.3 (algorithmic two-part code). Let ν be computable with decidable classes, and let K be prefix complexity. For every prefix p of length n ,

$$\begin{aligned} K(p) &\leq K(C_n(p) | n) + \lceil \log_2 \frac{\nu[C_n(p)]}{\nu[p]} \rceil + K(n) + O(1), \\ K(C_n(p) | n) &\leq \log_2 |P_n| + 2 \log_2 \log_2 |P_n| + O(1). \end{aligned}$$

If ξ is ν -Martin-Löf random, then by Levin–Schnorr there is c_ξ with $K(\xi_{1:n}) \geq -\log_2 \nu[\xi_{1:n}] - c_\xi$. Hence

$$K(C_n(\xi) | n) \geq -\log_2 \nu[C_n(\xi)] - K(n) - c_\xi - O(1).$$

Along random sequences the class label is incompressible relative to its own probability. The two-part code through classes is optimal up to the model deficiency $\delta_n(\xi) = K(C_n(\xi) | n) + \log_2 \nu[C_n(\xi)]$, which is bounded below by $-K(n) - c_\xi - O(1)$.

Proof. Given n and C , enumerate C with the computable weights ρ_C and use a Shannon–Fano–Elias code for p . Index the classes in a computable order. Combine with Levin–Schnorr. $\square$

Remark 2.4 (sophistication profile). Finite rank ($|P_n| \leq N$) means the model costs $O(\log N + \log n)$ bits per prefix: low sophistication. Example: golden mean = Fibonacci, two classes. Generic neural gains give singleton classes, where the model is the data: maximal sophistication, nothing strippable. Note I’s relaxation, with cost $I_\nu(P; Z | C)$ and at most N classes, trades model complexity against discarded predictive information. It is the Shannon analogue of the Kolmogorov structure function $h_x(\alpha) = \min\{\log |S| : x \in S, K(S) \leq \alpha\}$, and the rate–distortion problem $R(N)$ of note I is its neural version.

Theorem 2.5 (readout compression; randomness the network fails to strip). Let $H : X \rightarrow \mathbb{R}_{\geq 0}^k$ be an integrable network output on source sequences with source law μ . Put $S = \|H\|_1$, $M = \mu(S)$, direction $U = H/S$, and gain-weighted law $\nu = S\mu/M$. Fix any partition of level- n prefixes into classes C (coherent or not), and let q_p be the ν -conditional law of U given prefix p . Pick one representative $r_C \sim \rho_C$ independently per class. Then for every readout $u_j \in [0, 1]$,

$$\mathbb{E}_r \left| \mathbb{E}_\mu H_j - M \sum_C \nu[C] q_{r_C}(u_j) \right| \leq M \sqrt{\frac{1}{2} I_\nu(P; U | C)}.$$

If the classes are output-continuation classes ($q_p = q_q$ whenever p, q share a class), the error is 0. The 2^n -term prefix integral then collapses exactly to $|P_n|$ class integrals: only $\log_2 |P_n|$ of the first n seed bits are structure.

Proof. $\mathbb{E}_\mu H_j = M \mathbb{E}_\nu U_j = M \sum_C \nu[C] \bar{q}_C(u_j)$ with $\bar{q}_C = \sum_{p \in C} \rho_C(p) q_p$ (tower property). The error is at most $M \sum_C \nu[C] \mathbb{E}_{r_C} \text{TV}(q_{r_C}, \bar{q}_C)$. By Pinsker this is at most $M \sum_p \nu[p] \sqrt{D(q_p \| \bar{q}_C)/2}$, and by Jensen at most $M \sqrt{\sum_p \nu[p] D(q_p \| \bar{q}_C)/2}$. The last sum is $I_\nu(P; U | C)$. $\square$

3 Depth strips input randomness at a critical rate

Proposition 3.1 (structure fraction of the output). *Let $\kappa(c) = \sum_r \kappa_r c^r = \mathbb{E}_W[F_j(\theta)F_j(\theta')]$ for unit θ, θ' with $\langle \theta, \theta' \rangle = c$ (all $\kappa_r \geq 0$; note I), and let $m_r = \mathbb{E}_{\sigma \otimes \sigma} \langle \theta, \theta' \rangle^r$. Then*

$$\mathbb{E}_W \mathbb{E}_\sigma F_j^2 = \kappa(1), \quad \mathbb{E}_W (\mathbb{E}_\sigma F_j)^2 = \sum_r \kappa_r m_r = \kappa_0 + \sum_{r \geq 2 \text{ even}} \kappa_r m_r.$$

So the weight-averaged randomness fraction $\mathbb{E}_W \text{Var}_\sigma F_j / \mathbb{E}_W \mathbb{E}_\sigma F_j^2$ equals $1 - \sum_r \kappa_r m_r / \kappa(1)$. At infinite width ($\kappa \rightarrow \frac{1}{n} \varrho^{\circ L}$, $m_r = O(n^{-r/2})$) this tends to

$$P(Z_L > 0) \sim \frac{9\pi^2}{2L^2} \quad (\text{exactly } 0.0705 \text{ at } L = 16).$$

Proof. Expand κ and use $\mathbb{E} \langle \theta, \theta' \rangle^r = m_r$ (zero for odd r). The limit follows from note I (NNGP limit, Slack’s theorem for critical Galton–Watson processes with stable-3/2 offspring tail). $\square$

Proposition 3.2 (frozen address bits). *At infinite width, the layer- ℓ pre-activations of two independent inputs are jointly Gaussian with correlation $\varrho^{\circ(\ell-1)}(\langle \theta, \theta' \rangle)$ (NNGP recursion). So the activation bit of a layer- ℓ neuron disagrees with probability*

$$\frac{1}{\pi} \mathbb{E} \arccos(\varrho^{\circ(\ell-1)}(\langle \theta, \theta' \rangle)) \rightarrow \frac{\arccos(\varrho^{\circ(\ell-1)}(0))}{\pi} \approx \frac{3}{\ell}$$

(Sheppard’s formula, plus $1 - \varrho^{\circ \ell}(0) \sim 9\pi^2/(2\ell^2)$). The address $\pi(\theta)$ is thus a stretched seed whose fresh pairwise randomness per bit decays like $3/\ell$. The rest is structure fixed by the weights.

layer ℓ	1	2	4	8	12	16
disagreement, predicted	0.500	0.397	0.293	0.200	0.154	0.126
disagreement, observed ($n = 512$)	0.499	0.396	0.289	0.205	0.157	0.120
bits constant on all 4000 inputs	0.000	0.000	0.006	0.086	0.195	0.305

The observed final-layer randomness fraction at $n = 512$, $L = 16$ is 0.090, against the infinite-width prediction 0.0705.

Remark 3.3 (critical, not gapped). Note I’s Cantor heat strips fine levels doubly exponentially (spectral gap λ_1). That is why a $\log \log(1/\varepsilon)$ seed fools it. Depth strips input randomness only polynomially, because He initialization is exactly the critical point of the branching process. The surviving randomness is also spread over a heavy tail of high moment orders: the ReLU kink gives offspring tail $\sim r^{-5/2}$. The next section shows that this tail is exactly what no short seed can fake.

4 Inputs as seeds: identity, lower bound, near-optimal design

For a probability measure $\beta = \sum_i w_i \delta_{d_i}$ on S^{n-1} , define the seed estimator $\hat{f} = c_n \sum_i w_i F(d_i)$ and its moment defects $\Delta_r = \sum_i w_i d_i^{\otimes r} - \mathbb{E}_\sigma \theta^{\otimes r}$.

Theorem 4.1 (exact seed identity). *Condition on $W_2, \dots, W_L$ and average over the first-layer weights only. Then for every neuron j ,*

$$\mathbb{E}_{W_1}(\hat{f}_j - f_j)^2 = c_n^2 \sum_{r \geq 0} \kappa_r^{(j)} \|\Delta_r\|_F^2, \quad \|\Delta_r\|_F^2 = \sum_{i,k} w_i w_k \langle d_i, d_k \rangle^r - m_r,$$

where $\kappa^{(j)}(c) = \mathbb{E}_{W_1}[F_j(\theta)F_j(\theta')]$ for $\langle \theta, \theta' \rangle = c$ has nonnegative Taylor coefficients. Random seed sets are handled by taking expectations.

Proof. Write $F_j(x) = G(W_1 x)$ with G fixed. For unit a, b , the rows of $(W_1 a, W_1 b)$ are i.i.d. bivariate normal with correlation $\langle a, b \rangle$, so $\mathbb{E}_{W_1} G(W_1 a) G(W_1 b) = \kappa(\langle a, b \rangle)$. The coefficients are nonnegative by multivariate Mehler (note I). Expand $(\hat{f} - f)^2$ with (1) and use $\langle a, b \rangle^r = \langle a^{\otimes r}, b^{\otimes r} \rangle$. The cross term is $\sum_i w_i \mathbb{E}_\theta \langle d_i, \theta \rangle^r = m_r$ by rotation invariance. Absolute convergence holds because $\sum_r \kappa_r = \kappa(1) < \infty$. $\square$

Corollary 4.2 (standard seeds). *With M points:*

- *i.i.d. uniform:* $\mathbb{E} \|\Delta_r\|^2 = (1 - m_r)/M$ for $r \geq 1$;
- *antithetic pairs $\pm d$:* all odd r vanish, and even r behave like $M/2$ i.i.d. points;
- *spherical t -designs:* all $r \leq t$ vanish;
- *K independent Haar-rotated cross-polytopes $\{\pm O e_i\}$ ($M = 2nK$, a 3-design):* $\Delta_r = 0$ for $r \leq 3$ and all odd r , and exactly $\mathbb{E} \|\Delta_r\|^2 = \frac{1}{K}(\frac{1}{n} - m_r)$ for even $r \geq 4$.

Therefore

$$\frac{MSE_{iid}}{MSE_{frames}} = \frac{\sum_{r \geq 1} \kappa_r (1 - m_r)}{2 \sum_{r \geq 4 \text{ even}} \kappa_r (1 - nm_r)} \xrightarrow{n \rightarrow \infty} \frac{P(Z_L > 0)}{2 P(Z_L \in 2\mathbb{N}, Z_L \geq 4)}.$$

Theorem 4.3 (no seed beats the even high orders). *For every finitely supported probability β on S^{n-1} with nonnegative weights and at most M atoms,*

$$\mathbb{E}_{W_1}(\hat{f}_j - f_j)^2 \geq c_n^2 \sum_{r \geq 2 \text{ even}} \kappa_r^{(j)} \left(\frac{1}{M} - m_r \right)_+.$$

For $n \leq M \ll n^2$ the $r = 2$ term is vacuous and the bound is $\approx c_n^2 \sum_{r \geq 4 \text{ even}} \kappa_r / M$. Cross-polytope frames achieve $c_n^2 \sum_{r \geq 4 \text{ even}} \kappa_r (\frac{2}{M} - \frac{m_r}{K})$, which is **within a factor 2 of optimal**. Consequently no nonnegatively weighted seed set of size $M \ll n^2$ improves on i.i.d. sampling by more than

$$\frac{P(Z_L > 0)}{P(Z_L \in 2\mathbb{N}, Z_L \geq 4)} \approx 8.6, 4.8, 3.3 \quad (L = 4, 8, 16).$$

Proof. For even r every term $\langle d_i, d_k \rangle^r$ is nonnegative, so $\sum_{i,k} w_i w_k \langle d_i, d_k \rangle^r \geq \sum_i w_i^2 \geq 1/M$. Then apply Theorem 4.1, dropping the odd terms (which are ≥ 0). For frames, see Corollary 4.2. $\square$

shape	$P(Z_L > 0)$	pred. antithetic	obs. antithetic	pred. frames	obs. frames
$n = 256, L = 4$	0.319	1.31	1.36	4.33	4.29
$n = 256, L = 8$	0.166	1.18	1.20	2.42	2.64
$n = 256, L = 16$	0.0705	1.10	1.23	1.66	2.17
$n = 1024, L = 16$	0.0705	1.10	1.08	1.66	1.73

Gains are MSE ratios at equal numbers of forward passes ($M = 2nK$, $K = 4$), against an independent high-precision reference (300 frames at width 256, 120 frames at width 1024; the reference noise inflates the measured MSEs by about 1–3%). At large L/n the observed gains exceed the infinite-width prediction, which suggests finite-width κ_r put more mass on low orders.

Remark 4.4 (status and use). Orthogonal and antithetic sampling are known variance-reduction devices (orthogonal random features; orthogonal Monte Carlo). What is new here, as far as we know, is the exact error identity, the factor-2 optimality of frames among all nonnegative seed sets, and the Galton–Watson prediction of the achievable gain. A Haar rotation costs $O(n^3)$, negligible next to the $2n$ forward passes of a frame. Frames are therefore a free drop-in replacement for i.i.d. inputs in any sampling component. The ceiling (about $3.3\times$ at the challenge depth) says that beating sampling by more requires estimating *structure* directly (mechanistic methods), not better seeds.

5 The network’s own tiling of the input

Definition 5.1. For a unit input θ , let $\pi_\ell(\theta) \in \{0,1\}^n$ be the layer- ℓ activation pattern and $\pi(\theta) = (\pi_1, \dots, \pi_L)$ the *address*. A depth- ℓ *tile* is a nonempty set $\{\theta : \pi_{1..\ell}(\theta) = a\}$, a spherical polyhedral cell (the first layer gives a central hyperplane arrangement, and deeper layers give bent arrangements). Tiles refine with depth, giving a rooted tree, the network’s Michon tree, whose boundary points are addresses.

Proposition 5.2 (exact stripping inside a tile). *On a depth- L tile τ , $F(x) = A_\tau x$ with $A_\tau = D_L W_L \cdots D_1 W_1$ and $D_\ell = \text{diag}(\pi_\ell)$. Hence*

$$\mathbb{E}_\gamma F = \sum_{\tau} A_\tau \mathbb{E}_\gamma[X \mathbf{1}_\tau(X)], \quad \mathbb{E}[F \mid \tau] = A_\tau \mathbb{E}[X \mid \tau].$$

The position of the input inside its tile is randomness the network strips exactly: only the tile’s Gaussian barycentre survives. For any depth ℓ , $\mathbb{E}[F \mid \mathcal{T}_\ell]$ is a Rao–Blackwellization with the same mean and smaller variance.

Proposition 5.3 (the Cantor machinery lives on the network’s tiles). *The σ -algebras $\mathcal{T}_\ell$ generated by depth- ℓ tiles are nested, so $\mathbb{E}_\ell := \mathbb{E}[\cdot \mid \mathcal{T}_\ell]$ satisfy $\mathbb{E}_\ell \mathbb{E}_m = \mathbb{E}_{\min(\ell, m)}$. For any $0 = \lambda_0 < \lambda_1 < \cdots$, the operator*

$$T_t^{\text{net}} = \sum_{\ell} p_\ell(t) \mathbb{E}_\ell$$

is a σ -symmetric Markov semigroup on $L^2(S^{n-1})$ with gap λ_1 (the nested-expectation construction of the continuation note, with tiles in place of cylinders). It preserves the readout $\mathbb{E}_\sigma F$ and progressively replaces F by tile averages: it strips the randomness in the tile by diffusion. Continuation laws are the conditional laws of the future address and output given a tile. Their classes are the network’s causal states, and Theorem 2.5 applies verbatim with tiles as prefixes.

Remark 5.4 (where the input sits in the arrow algebra). This is the abstract role of the input. It is a point of the network-induced hull, the space of addresses with the measure induced by σ , read through its tile address. Arrows join tiles with equal continuation laws, the centre records the network’s causal state, and the readout is a state applied to the diffusion. Proposition 3.2 says deep address bits are increasingly frozen, so continuation laws of different shallow tiles converge with depth and classes merge. Proposition 3.1 says this convergence is only polynomial (critical).

Conjecture 5.5 (sophistication decreases with depth, critically). For the network’s tile hierarchy with σ , let $N_\ell(\varepsilon)$ be the least number of coherent classes at depth ℓ for which Theorem 2.5 gives error $\leq \varepsilon$ on all output readouts. Then $N_\ell(\varepsilon)$ is eventually non-increasing in the remaining depth $L - \ell$. In the infinite-depth limit the address process has trivial tail (unique KMS_1 state in the sense of note I’s KMS theorem), but there is no geometric Birkhoff rate: forgetting is polynomial, with the Galton–Watson survival law $\sim 9\pi^2/(2(L - \ell)^2)$.

6 What this says about estimation

1. *Seeds.* The input’s randomness reaches the output only through Galton–Watson-weighted moment orders. Frames kill orders ≤ 3 for free, and nothing kills even orders ≥ 4 . Gain over Monte Carlo is $\approx 1.7\text{--}2.2\times$ at depth 16 (more at small L/n), with a hard ceiling of about $3.3\times$ for any seed design.
2. *Structure.* The remaining factor must come from computing structure: tile barycentres, class integrals (Theorem 2.5) or cumulants. Proposition 3.1 says the structure is 93% of the output’s second moment at depth 16. This is why mechanistic estimates are plausible at depth, and also why their residual error lives in the high-order tail.
3. *Synthesis target.* A hybrid that computes structure mechanistically and samples only the residual randomness with frames. Whether the residual’s Galton–Watson tail is small enough at 1024×16 is the next quantitative question.

A Numerical checks

Scripts are in `code/`: `seeds.py` (width 256) and `seeds32.py` (challenge shape; float32) compare i.i.d., antithetic and frame seeds, with three networks at width 256 and two at width 1024. `stripping.py` measures pattern disagreement, frozen bits and the randomness fraction. Galton–Watson spectra come from note I’s `spectrum.py`.